

UNIVERSAL GATES

- Although we can implement **any circuit** with **AND/OR/NOT**, we can also implement **any circuit** with **only NAND** or **NOR** gates.
- We might want to do this because of technology considerations, that is, these gates might be cheaper to implement in silicon or they might be the only type of gates we have available.
- Since we can always use only NAND or NOR gates, these gates are sometimes called **universal gates**.
- The “trick” (if you want to call it that) is to see that we can implement the three basic gates (AND, OR, NOT) in terms of NAND or NOR gates.

Universal Property of NAND and NOR gates:

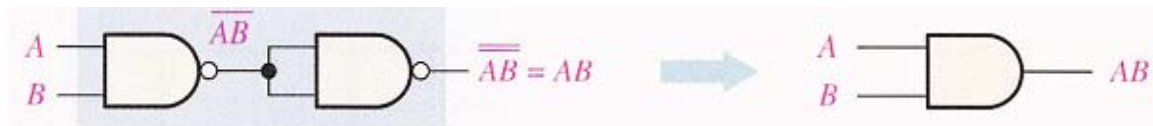
The NAND gate as a Universal logic gate:

- 1) One NAND gate used as an inverter (NOT gate):

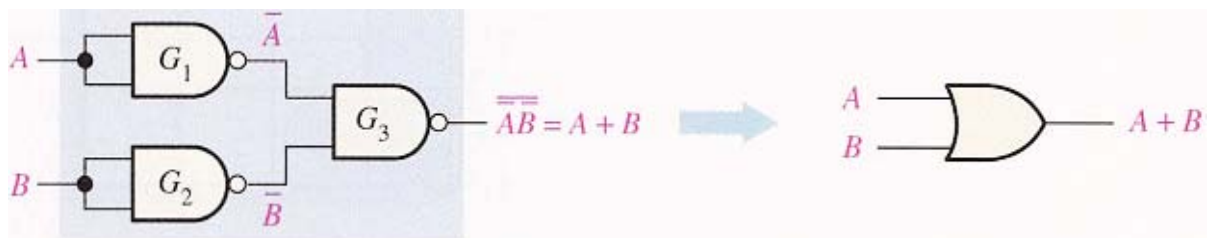


$$f = \overline{x \bullet x} = \bar{x}$$

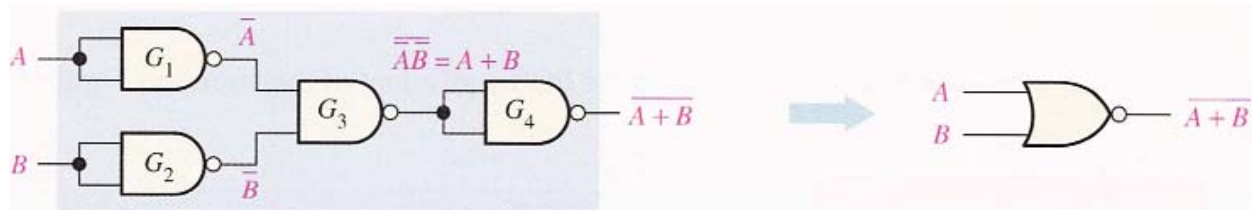
- 2) Two NAND gates used as an AND gate:



- 3) Three NAND gates used as an OR gate:



- 4) Four NAND gates used as a NOR gate:



The NOR gate as a Universal logic gate:

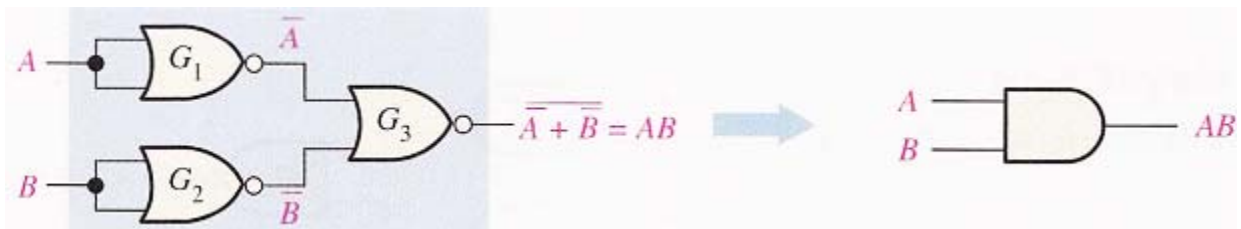
- 1) One NOR gate used as an inverter (NOT gate):



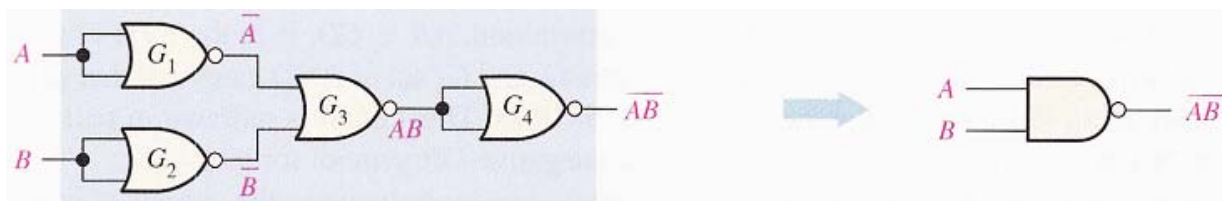
- 2) Two NOR gates used as an OR gate:



- 3) Three NOR gates used as an AND gate:



- 4) Four NOR gates used as a ~~NOR gate~~ NAND Gate



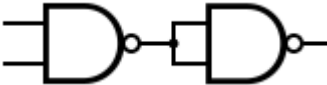
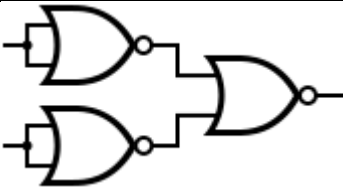
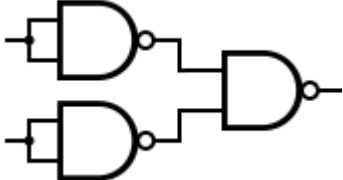
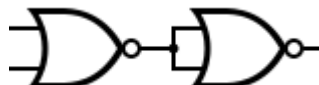
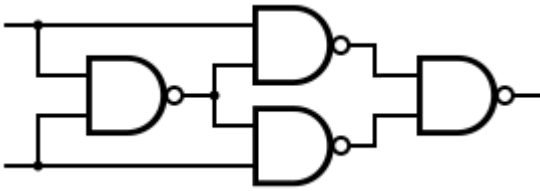
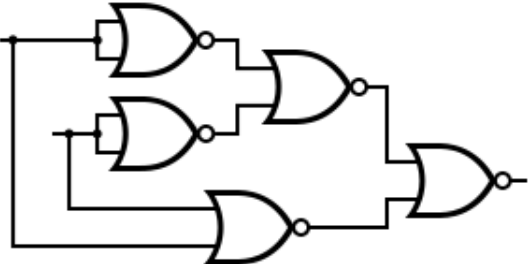
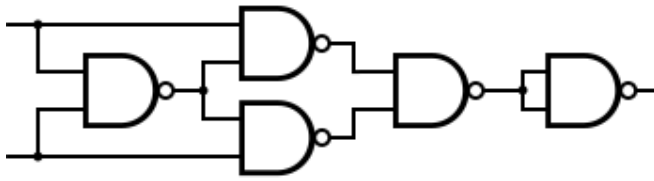
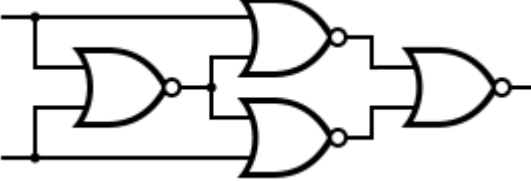
Implementing ANY circuit using Universal gates:

The ability to implement NOT/AND/OR with NAND or NOR means we can implement **any circuit** using only with **NANDs** or **NORs** by **only replacing** the AND, OR and NOT gates with their equivalent NAND or NOR implementations.

Procedure:

- 1) First construct the logic expression using basic gates (AND/OR/NOT),
- 2) Replace each basic gate with its equivalent universal gate implementation,
- 3) Cancel two consecutive NOT gates (according to Boolean algebra),
- 4) Redraw the final circuit.

Table 1: Logic gate equivalence with universal gate implementation

	NAND	NOR
NOT		
AND		
OR		
XOR		
XNOR		

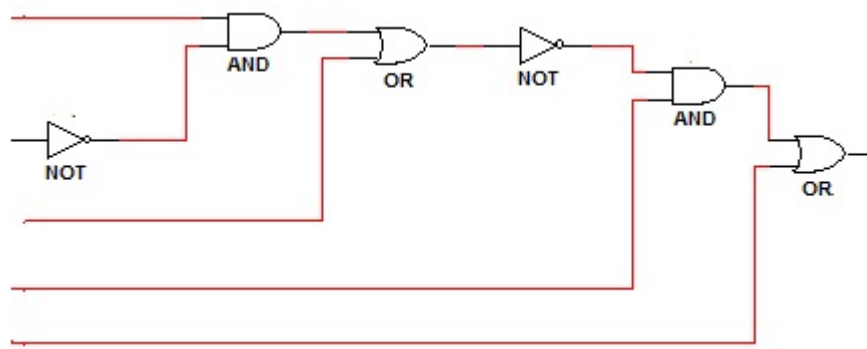
Example: Implement the following expression with (a) NAND gates ONLY, (b) NOR gates ONLY.

$$Y = (AB' + C)'D + E$$

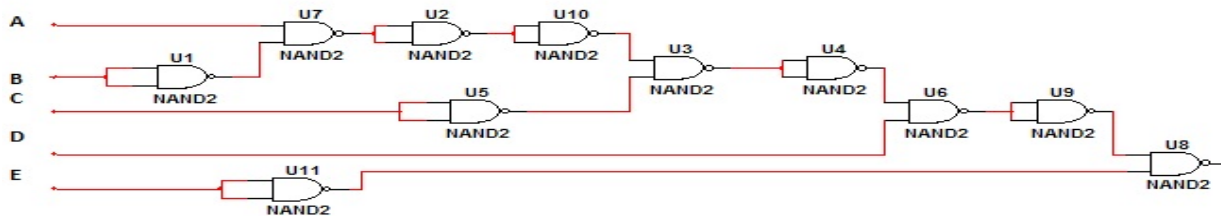
Solution:

(a) Implementation using NAND gates ONLY:

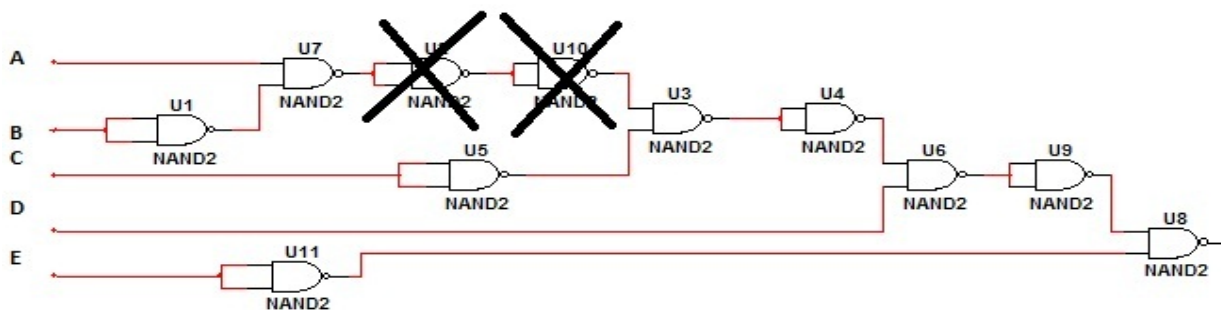
(i) Implementation with Basic gates:



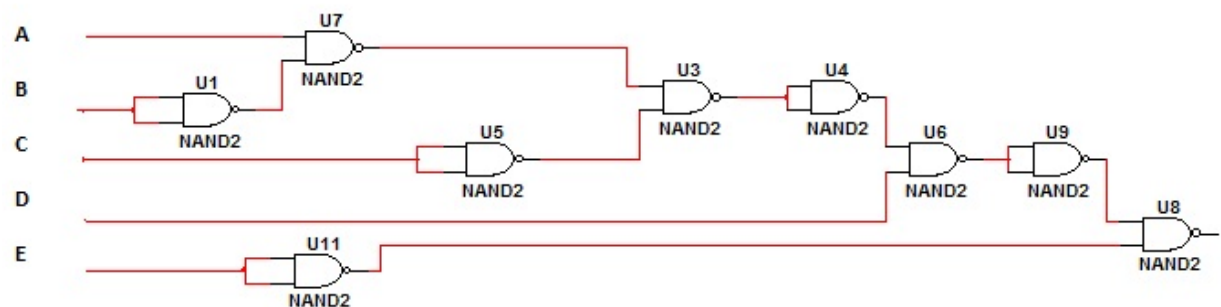
(ii) Replace each basic gate with its equivalent NAND gate implementation from Table 1.



(iii) Cancel two consecutive NOT equivalent gates (according to Boolean algebra).

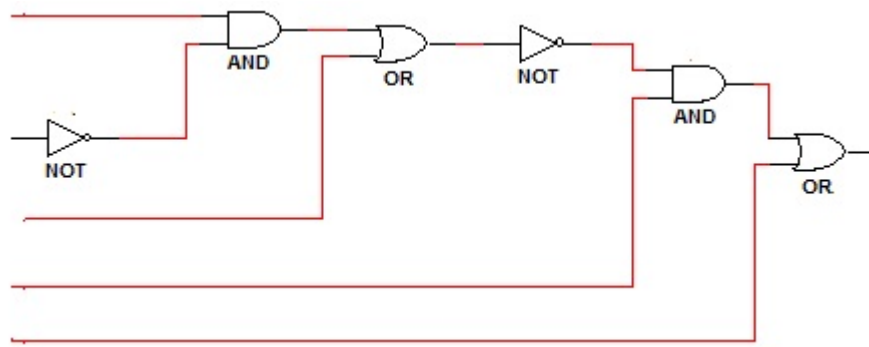


(iv) Redraw the final circuit.

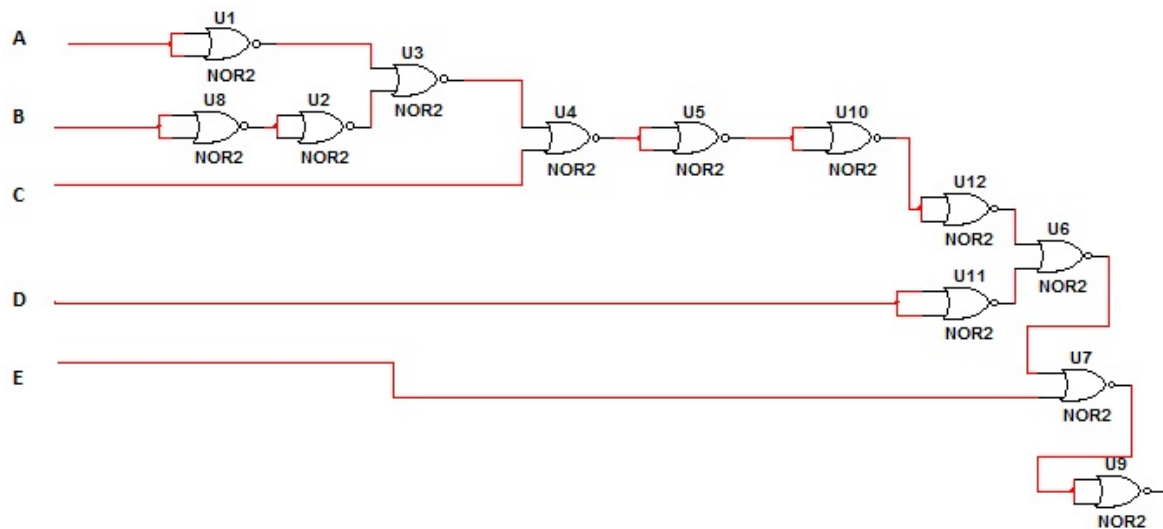


(b) Implementation using NOR gates ONLY:

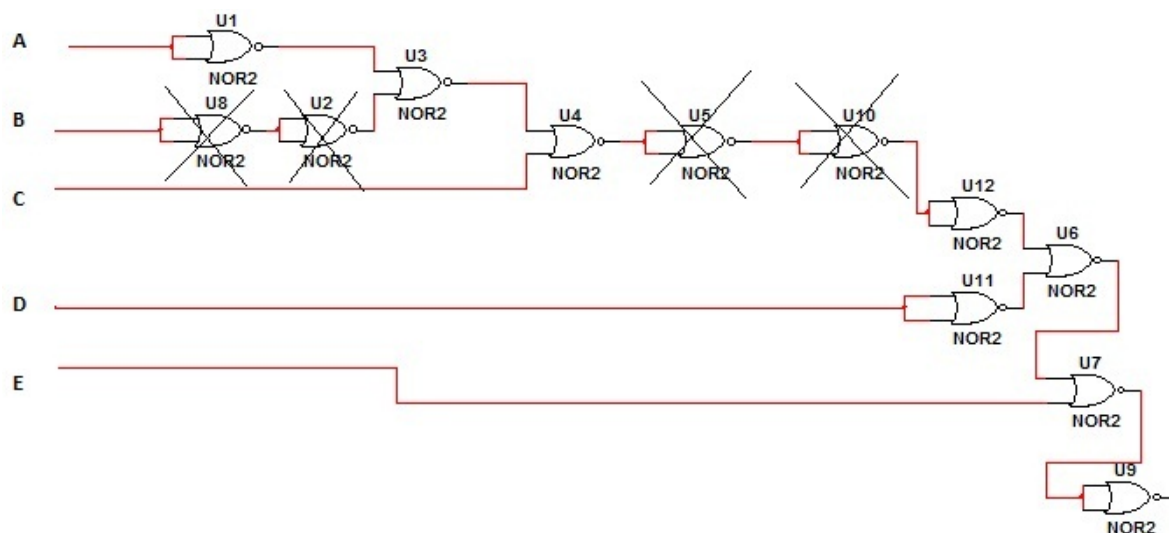
(i) Implementation with Basic gates:



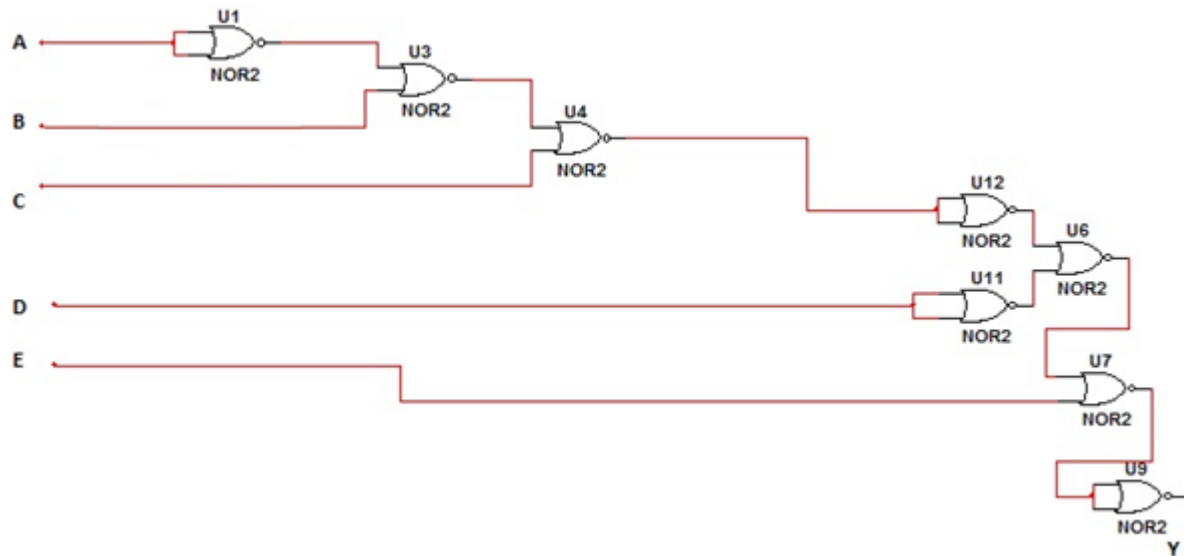
(ii) Replace each basic gate with its equivalent NOR gate implementation from Table 1.



(iii) Cancel two consecutive NOT equivalent gates (according to Boolean algebra).

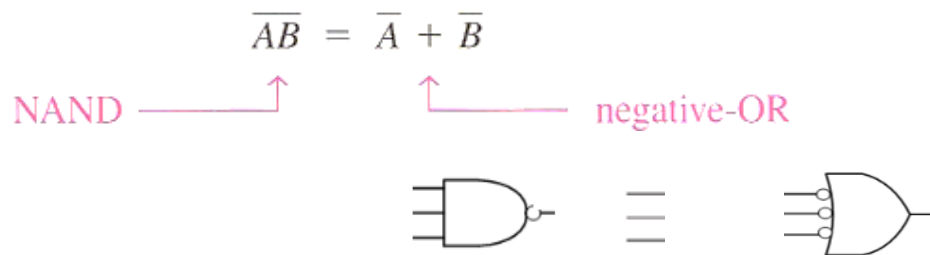


(iv) Redraw the final circuit.

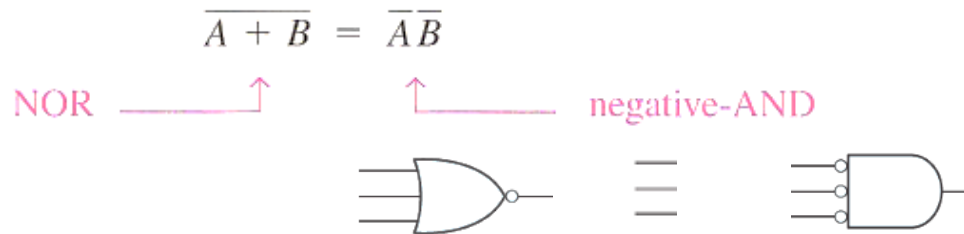


Dual Symbols:

NAND Logic: a NAND gate can function as either a NAND or a negative-OR

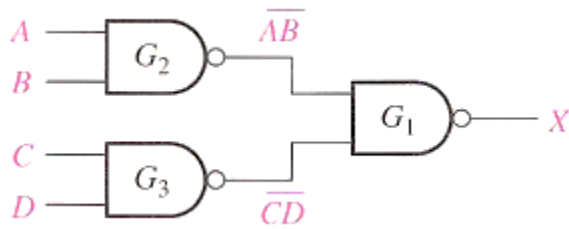


NOR Logic: a NOR gate can function as either a NOR or a negative-AND



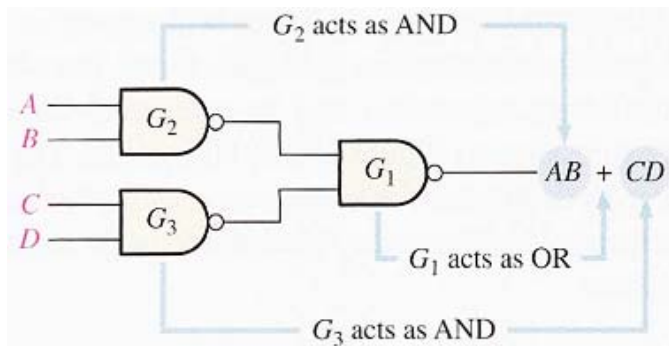
COMBINATIONAL LOGIC USING NAND AND NOR GATES

1) NAND gates:

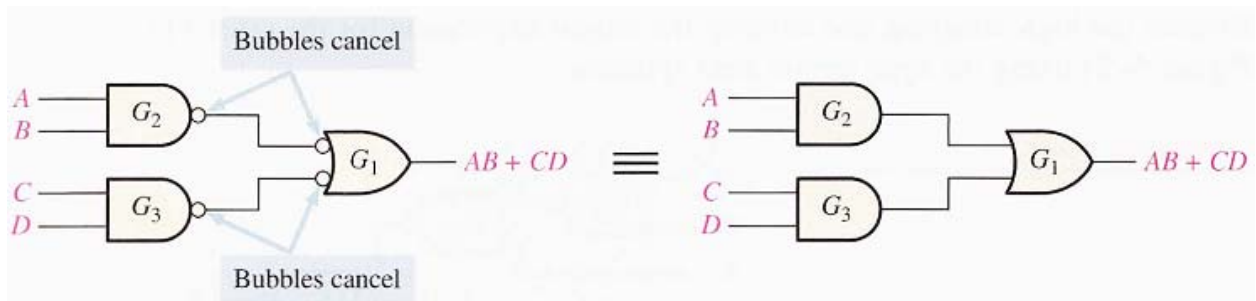


$$\begin{aligned}
 X &= \overline{(\overline{AB})(\overline{CD})} \\
 &= \overline{(\overline{A} + \overline{B})(\overline{C} + \overline{D})} \\
 &= \overline{(\overline{A} + \overline{B})} + \overline{(\overline{C} + \overline{D})} \\
 &= \overline{\overline{A}}\overline{\overline{B}} + \overline{\overline{C}}\overline{\overline{D}} \\
 &= AB + CD
 \end{aligned}$$

NAND Logic Diagrams Using Dual Symbols



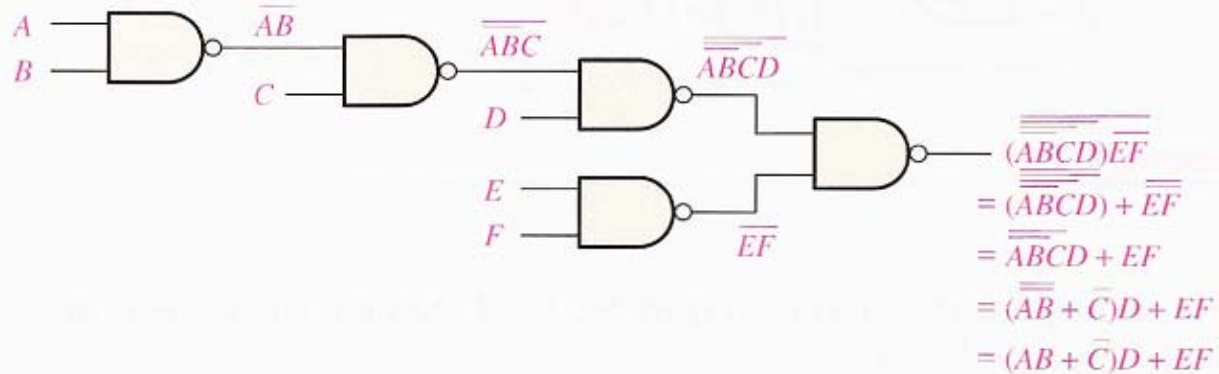
(a) Original NAND logic diagram showing effective gate operation relative to the output expression



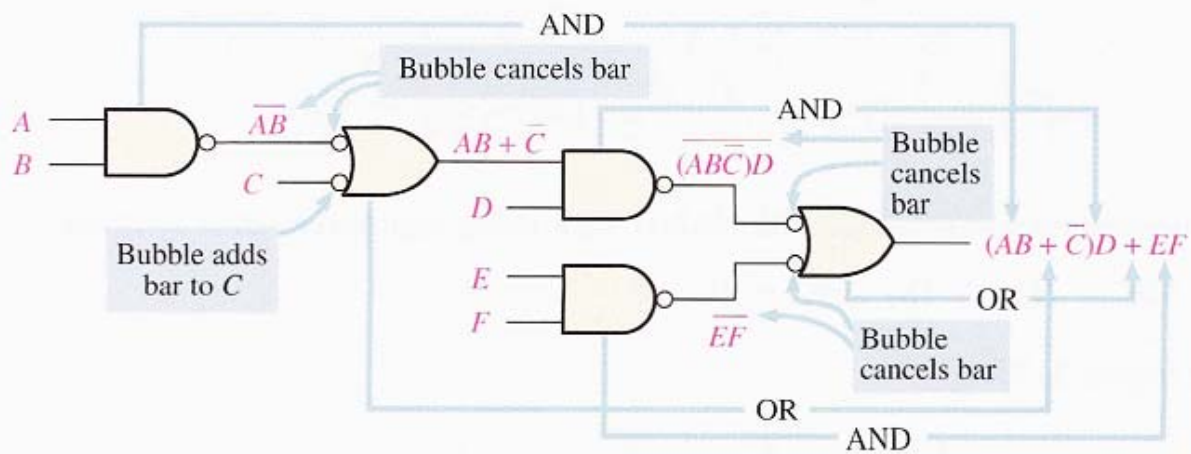
(b) Equivalent NAND/Negative-OR logic diagram

(c) AND-OR equivalent

Example:



(a) Several Boolean steps are required to arrive at final output expression.

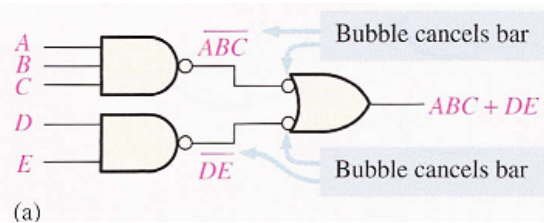


(b) Output expression can be obtained directly from the function of each gate symbol in the diagram.

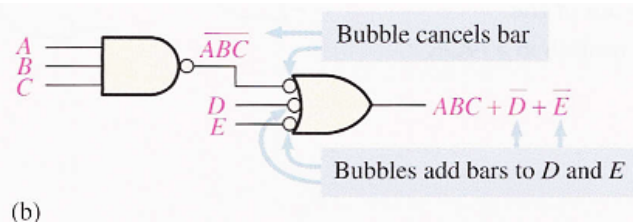
EXAMPLE

Implement each expression with NAND logic using appropriate dual symbols:

(a) $ABC + DE$ (b) $ABC + \overline{D} + \overline{E}$

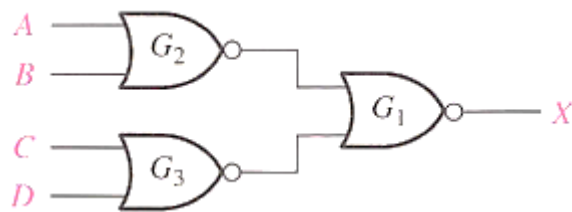


(a)



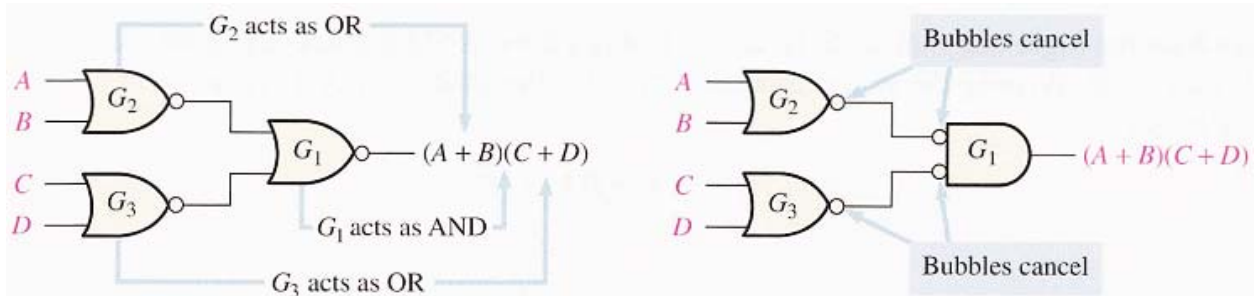
(b)

2) NOR gates:

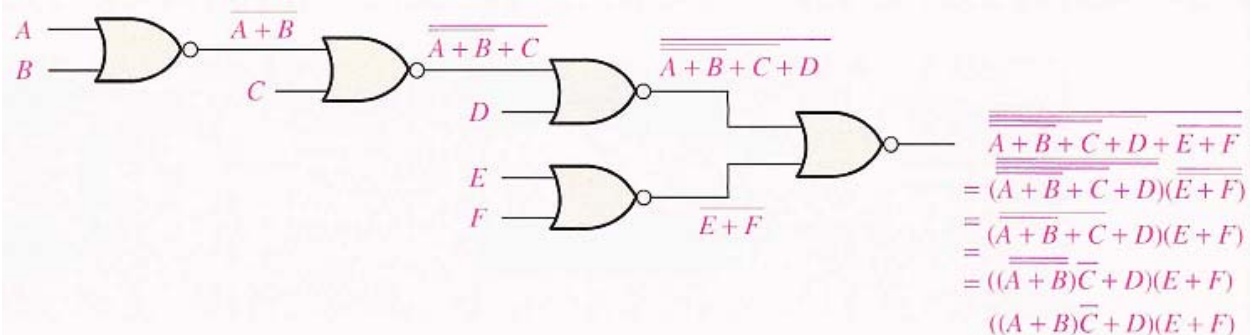


$$X = \overline{\overline{A + B + C + D}} = (\overline{\overline{A + B}})(\overline{\overline{C + D}}) = (A + B)(C + D)$$

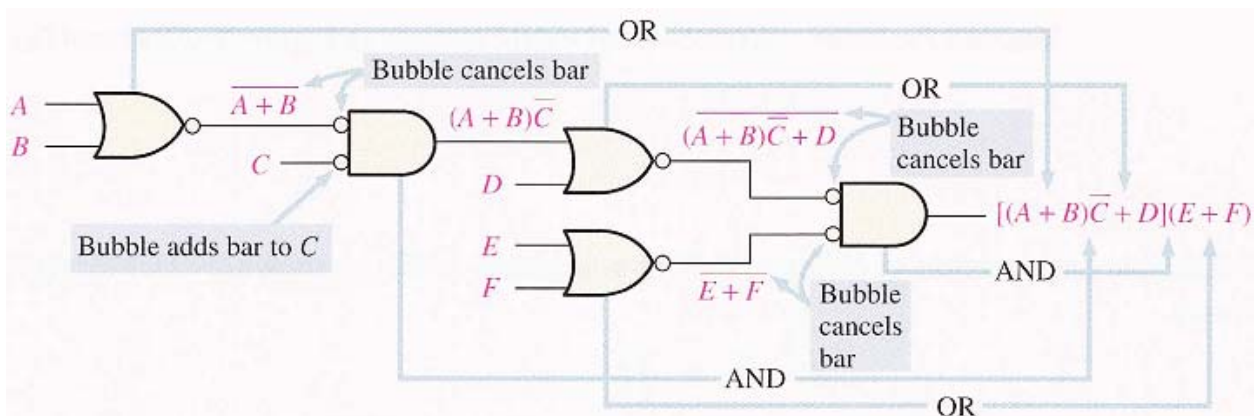
NOR Logic Diagram Using Dual Symbols



Example:



(a) Final output expression is obtained after several Boolean steps.



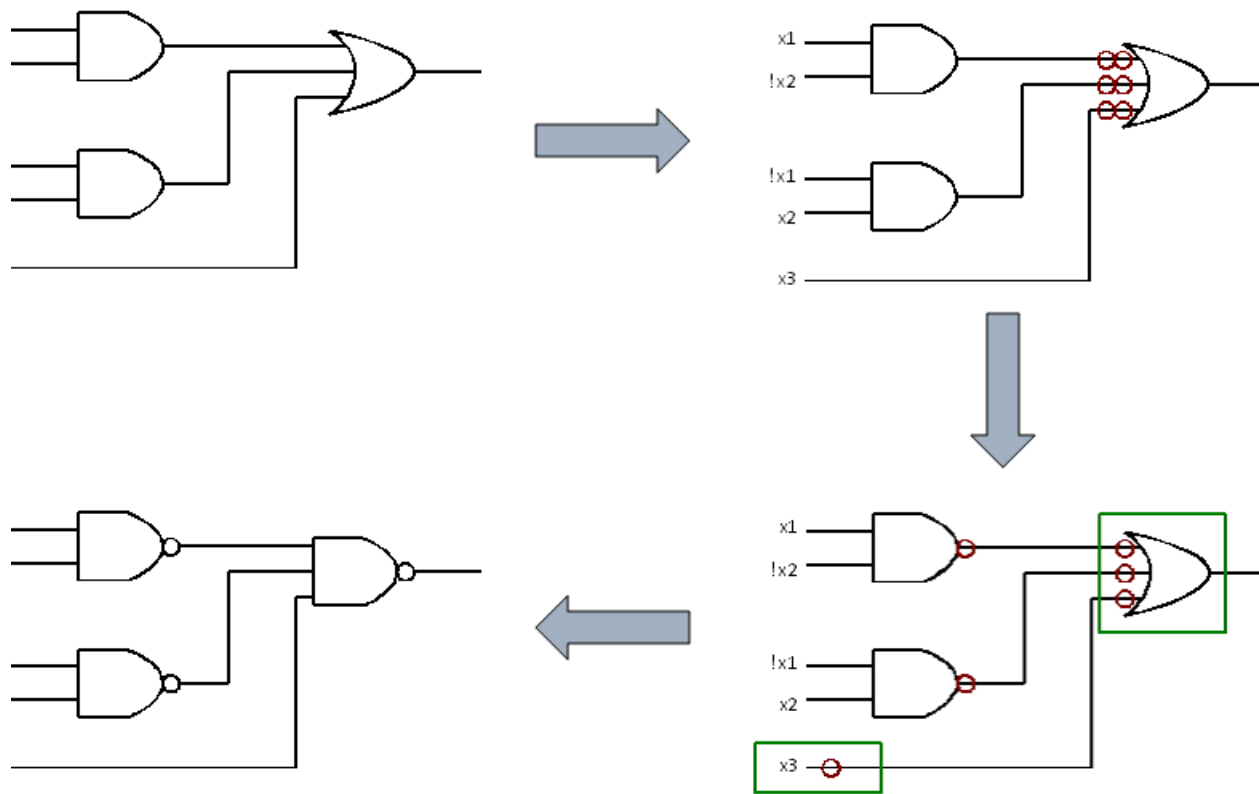
(b) Output expression can be obtained directly from the function of each gate symbol in the diagram.

Implementing ANY SOP circuit using NAND gates (graphically):

2 Level implementation

We can also make the conversion graphically. Again, given a 2-level SOP the procedure is straightforward:

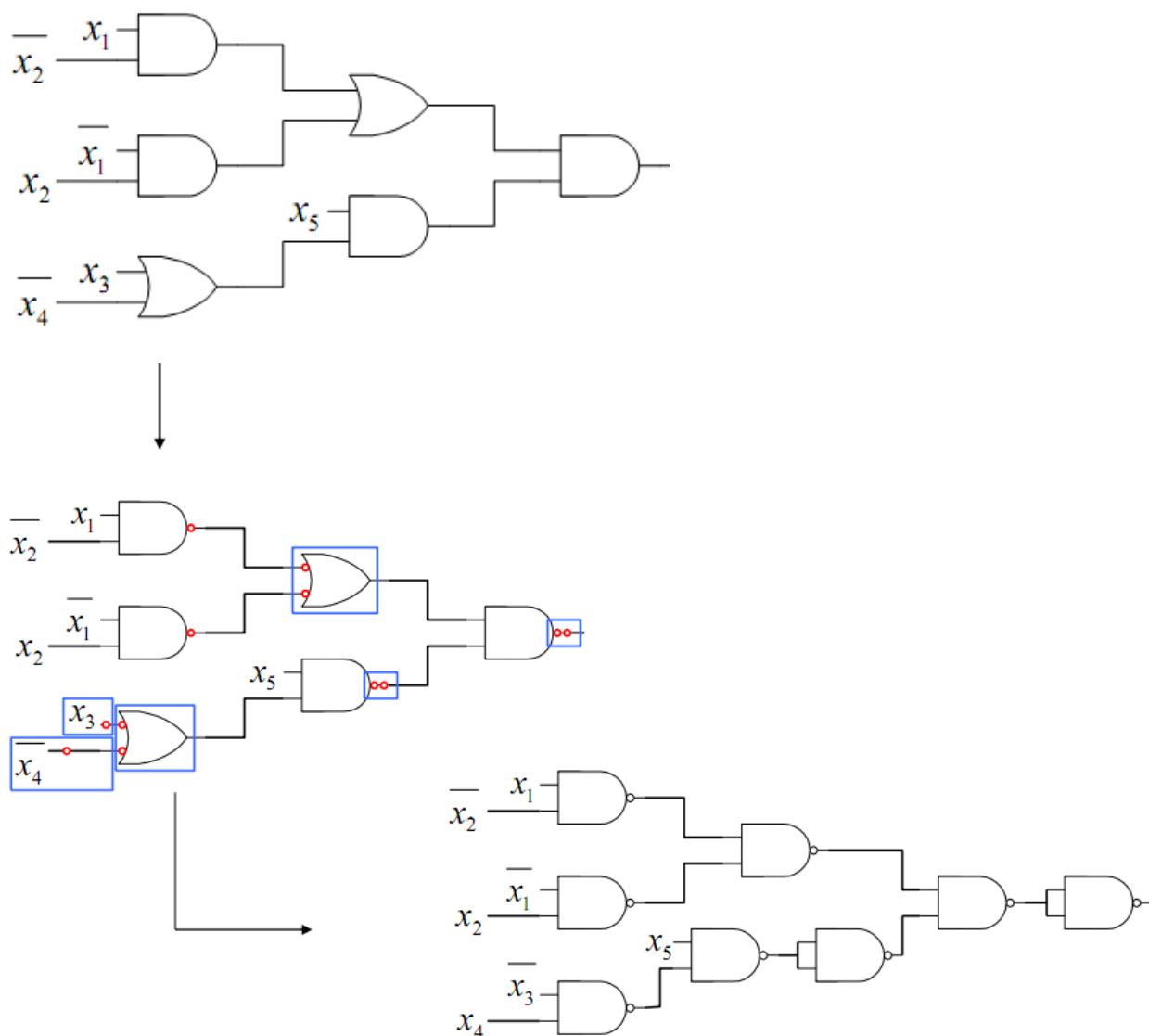
- Insert double inverters at input of OR gate (input will be either from an AND gate, or from an input literal).
- Convert OR gate plus one level of inverters to NAND gate.
- Convert AND-gates plus the other level of inverters to NAND gates.
- Some literals might get inverted.



Multi Level implementation

Circuits are not always in 2-level (or SOP) form. Conversion method is essentially the same:

- Insert double inverters before OR gates.
- Use one set of inverters plus the OR gates to convert to NAND.
- Use other inverts with AND gates to convert to NAND.
- Occasionally, might need to implement an inverter using NAND.



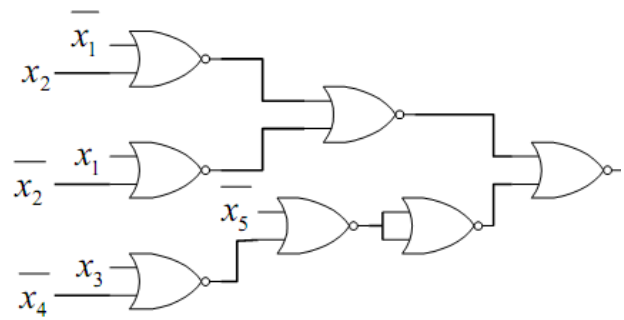
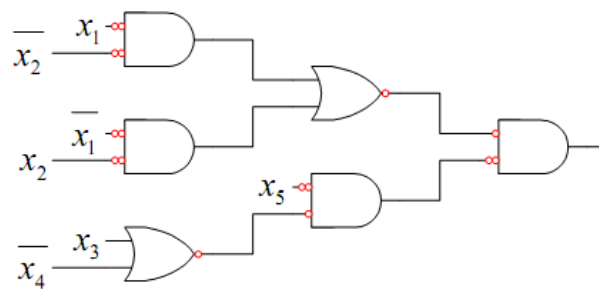
Notice the **extra** inverters...

Implementing ANY POS circuit using NOR gates (graphically):

2 Level implementation

Can also make the conversion graphically and it is straightforward:

- Insert double inverters at input of AND gate (input will be either from an OR gate, or from an input literal).
- Convert AND gate plus one level of inverters to NOR gate.
- Convert OR gates plus the other level of inverters to NOR gates.
- Some literals might get inverted.



Notice the **extra** inverters...